Political Competition Over Life and Death

Social Provision and Infant Mortality in India

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Abstract

We argue that economic inequality harms social provision to the poor, but that higher political competition can mitigate this effect. We test this hypothesis using a large redistricting of electoral boundaries in India. Higher economic inequality leads to more post-neonatal infant deaths, but only when political competition is weak. We assert that the effect on mortality operates via changes in social provision at the local level and confirm this for two different programs: Inequality leads to worse performance of public healthcare and weaker provision of the workfare program MGNREGA, but only in situations with little political competition.

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I Introduction

"Votes count, but resources decide" (Rokkan 1966, p. 105). If what Rokkan said is true, high economic inequality can distort the provision of social necessities to the poor, as it may empower the special interests of the rich, trigger rent-seeking and facilitate corruption. Yet, as votes also count, the interests of a large number of poor people might become more decisive the tighter the election. In a tight election, the incumbent seeking reelection cannot entirely neglect the needs of the poor. Therefore, a high level of political competition can fuel spending on public provisions and mitigate the adverse impact of economic inequality.

We test this hypothesis using fine-grained data from India. We use infant mortality as an outcome measure, because it is affected by public provision at the local level. Over the last three decades, there has been a clear improvement in child survival worldwide, and India is no exception (Hug, Sharrow and You 2019) – although there is a substantial variation. The worst-performing state, Madhya Pradesh, has an estimated infant mortality rate 14 times higher than the best-performing state, Mizoram (Duggal and Dilip 2022).[1] Infants still die from preventable diseases and injuries, highlighting the need for social provision. To test the hypothesis more directly, we also study two types of social provision for the poor: basic healthcare and employment. For healthcare, we focus on the Primary Health Centers (PHCs), which provide basic health products and services and constitute the cornerstone of the health system in rural India. For employment, we focus on the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA), the world's largest employment program, which guarantees 100 days of minimum-wage employment per year for each rural household.

To make causal inferences, we utilize the large redistricting of parliamentary constituencies in 2008, the so-called Delimitation. Our empirical strategy is to compare locations that were in the same constituency before the Delimitation but in different constituencies after the Delimitation. We implement this by regressing our outcome variables from the period after the redistricting on constituency-level inequality and political competition, controlling for pre-delimitation constituency fixed effects. This is like a natural experiment where locations are reallocated into constituencies with different levels of inequality and political competition.[2] We interact the fixed effects with identifiers at the district level, which is the most important administrative unit for social provision to the poor.

The main identifying assumption is that – accounting for the fixed effects – no factors but the changes in inequality and political competition at the constituency level affect our outcomes of interest. We believe the assumption is plausible, given how the Delimitation was implemented and that our identifying variation is at a fine geographical level. We test the assumption in several ways. First, we look for evidence of gerrymandering by investigating whether potentially influential politicians experienced an ex-ante more favorable redistricting than other incumbent politicians.[3] We do not find evidence of such favoritism. Second, we show that, for a wide range of observables, locations that were allocated to new constituencies do not differ from locations that were not. Third, we conduct a placebo exercise using outcomes from before the Delimitation, and show that our empirical setup does not predict these outcomes.

One remaining concern is that our explanatory variables – economic inequality and political competition – pick up the effects of other correlated constituency characteristics. We cannot exclude this possibility, but we consider it unlikely, as parliamentary constituencies do not *implement* local policies. Indeed, the only impact on local policies is through the influence of the members of parliament (which is the effect we are interested in). Differences in administrative and financial capacity – which are likely to be important predictors of differences in public goods provision – should play a limited role in our setup. These differences may matter at the level of districts, but we control for those through the fixed effects. The key remaining factors that could threaten the empirical setup are, therefore, characteristics of the constituencies that influence the decisions taken by the members of parliament. Such factors are not likely to be numerous. Still, to alleviate some of this concern, we run horse race regressions, where we interact our main variables of interest with a large set of constituency characteristics, such as poverty, urbanization and literacy, and show that our key results are robust to these more demanding specifications.

Our main finding is that economic inequality leads to i) more post-neonatal infant deaths,[4] ii) less functioning public health clinics, and iii) lower disbursement of wages through the workfare program – but only in situations with weak political competition. Where there are average levels of political competition, inequality has little impact on most of our outcomes. When the level of political competition is one standard deviation below the sample mean, a one standard deviation increase in inequality leads to a rise in post-neonatal infant mortality of 13% of the sample mean, a 0.15 standard deviations fall in our healthcare index, and a reduction of 6% of the sample mean in the amount of wages disbursed through MGNREGA. None of these effects are found in the placebo regressions.

Our conclusions are based on evidence that is derived from three independent datasets. In our view, this significantly strengthens the plausibility of our findings. The National Family and Health Survey (NFHS) is the data source for mortality. For PHCs we use the District Level Household and Facility Survey (DLHS), and data on MGNREGA was scraped from the MGNREGA Public Data Portal.

We interpret the mortality results as reduced form effects of inequality and political competition, which operate through public provision at the local level. Our healthcare and public work results are consistent with that interpretation, but we do not claim that the mortality results are only driven by these types of social provision. Other public services, such as clean water and sanitation, might be equally or even more important. Unfortunately, there is no sufficiently fine-grained data to conduct a similar analysis for these provisions.[5] Still, the results on PHCs and MGNREGA provide important evidence in support of our proposed mechanism.

Our investigation is focused on the interaction between economic inequality and political competition. The theoretical foundation for our approach – as we elaborate further in Section II – belongs to the literature on i) how inequality can empower special interests and ii) how political competition can empower the poor majority and reduce the influence of special interests, which in turn can improve policies directed at the poor.

Like Kudamatsu (2012) and Ross (2006), we emphasize the effect of democracy on health and public healthcare spending. Unlike most empirical studies, however, we explore changes in political competition at the *intensive* margin in a setting where democracy is already well established.[6] Also, few existing papers test the theoretical prediction that electoral competition increases voter welfare. This may be particularly relevant in low- and middle-income countries, where other institutions of checks and balances can be weak. In Sierra Leone, chiefdoms that had less political competition during colonial rule have worse development outcomes today (Acemoglu, Reed and Robinson 2014). In Brazil, electoral competition increases the supply of public goods (Arvate 2013). In India, economic outcomes are not always better in more competitive constituencies (Shaukat 2019), and contested incumbents can trade off less visible health centre capacity for more visible access to health centres (Kailthya and Kambhampati 2022). Therefore, it is important to explore the role of political competition for both social provision and direct health outcomes. Finally, we build on and add to studies that use the variation created by the Delimitation to investigate how politicians divert the allocation of public resources (see for example Bardhan et al. 2020; Jensenius and Chhibber 2023; Nath 2014).

In Section II, we describe the motivation for our empirical investigation and our contribution to the literature with a general discussion of the link between income inequality and social provision, and how it depends on political competition. In Section III, we provide details on the institutional background and the Delimitation. In Section IV, we discuss our identification, present the different data sources, define the key variables, and assess our empirical model. We discuss the results in Section V and conclude in Section VI.

II Theoretical Motivation

In India and other low-and middle-income countries, poor individuals lack access to essential services such as healthcare, clean water, sanitation, education, and employment opportunities. Local political leaders often face a dilemma between providing these public goods and catering to the specific needs of certain groups. Economic inequality further complicates this trade-off, making it challenging to provide adequate social support for the poor.

As discussed by Banerjee (1997), Baland and Platteau (1997), Bardhan and Mookherjee (2000) and Esteban and Ray (2006), unequal political influence can be a result of inequality empowering the special interests of the rich. Articulating collective demands requires coordination, which can be more difficult in large groups with low incomes than in smaller groups with high incomes. Under this constraint, participation in social activities is lower in more unequal societies (Alesina and La Ferrara 2000). The corresponding concentration of income among the rich in unequal societies does not create similar free-riding problems (Olson 1965), making the implementation of their preferred policies more likely. Such unequal influence is particularly relevant for health issues (Lynch et al. 2001; Judge, Mulligan and Benzeval 1998; Wagstaff 2003; Wilkinson 2002; Deaton 2003).

We combine the link from inequality to political influence with the counteracting effect of political competition, building on Barro (1973), Coate and Morris (1995), Besley, Persson and Sturm (2010) and Wittman (1989). In short, political pressure from the challengers can discipline the incumbent to implement social policies to attract voters.

The crucial decision for the incumbent is how to best allocate limited resources within the constituency in a way that mobilizes the necessary electoral support. As outlined below, this requires balancing the basic needs for public provisions with the specific needs of certain groups, accounting for the political benefits and the political opportunity costs.

II.A The Political Opportunity Costs

In practice, the temptation to divert resources away from public provision depends on how the incumbent perceives the personal and social gains of their alternative use. The opportunity costs include potential gains from illicit activities like corruption and bribery. Economic inequality plays a significant role, as wealthy individuals have the means to offer substantial bribes. Therefore, part of the opportunity cost of investing in public goods is forgoing bribes from those willing to pay them.

The costs are also influenced by legitimate lobbying from groups that signal strong demands for specific uses of the available resources. Even a well-intentioned politician, aiming to make rational decisions based on updated beliefs, may struggle to interpret lobbying signals correctly. Do the signals reflect genuine desirability or simply easy access to financial resources? As discussed by Esteban and Ray (2006), both high wealth and true economic desirability can be behind influential lobbying efforts that, in our case, can bias the perceived gains of preferential treatment.

Economic inequality exacerbates the issue. Richer groups have better access to financial resources and are, therefore, more likely to influence the incumbent's perception of gains from preferential treatment than other groups. Economic inequality increases illegitimate and legitimate influence activities, ultimately raising the political opportunity costs of investing in public good.

To decide the level of public provision, however, the incumbent must weigh these political opportunity costs of public goods against their political benefits.

II.B The Political Benefits

One important benefit of public goods provision for the incumbent is that they can improve their chance of being reelected. Confronted with political challengers, the incumbent can use social policies to capture votes.

A contest between a political incumbent and their challengers has some similarities to a contest in which there is rent seeking and a wasteful use of resources. An important distinction, however, is how electoral contests, contrary to rent-seeking contests, can discipline the incumbents and induce them to provide essential public goods to gain the support of economically disadvantaged voters.

This strategy is not equally gainful under all circumstances. When the incumbent is expected to either win or lose the election with a clear margin, the political benefits of increasing the provision of public goods are limited. However, the benefits are high when the chances of winning for either candidate take intermediate values.

In Online Appendix A, we present a stylized formalization of how political competition can prevent the misappropriation of funds for private interests and promote provision of public goods for everyone instead. The analysis highlights the role of the endogenous probability s of reelecting the incumbent. He is more likely to pander to special interests when, regardless of his actions, he is almost certain to be reelected (s close to 1) or almost certain to lose (s close to zero).

We find that the expected political benefits of providing public goods are proportional to $s(1-s)$ – a term close to the widely used measure of political competition, namely one minus the Herfindahl-Hirschman index,[7] which we use in the empirical investigation. It also turns out that the value of the expected political benefits increases with our measure of political competition and reaches its highest level when $s = 1/2$.

In other words, for any given level of the political opportunity costs, the maximum level of public provision is associated with the most equal levels of political competition. Therefore, strong political competition is essential for the effective provision of public goods.

There may also be a reinforcing effect between lobbying and political competition.

When the competition is expected to be tight, those who lobby for preferential treatment can anticipate that the impact of their lobbying efforts will be modest, as the incumbent would invest in public goods to maintain power. As a result, groups may lobby less, further reducing the perceived opportunity costs for the incumbent and promoting even greater provision of public goods.

II.C The Race Between Economic Inequality and Political Competition

Economic inequality tends to increase the opportunity costs of providing public goods, and political competition tends to reduce the political benefits. Higher opportunity costs lead to lower social provision, and more political competition leads to higher social pro- vision (shown as Results 1 and 2 in Online Appendix A).

This means that even though a tight political contest may increase social provision for the poor, the political consequences of high economic inequality can work in the opposite direction. Can political competition fuel public goods provision so much that it eliminates the bad consequences of inequality?

It follows from our discussion that political competition can reduce the impact of high political opportunity costs. It also follows that a particularly fierce political competition can induce the incumbent to allocate all available resources to public goods provision (a sufficient condition is shown in Online Appendix A). Then, of course, somewhat higher political opportunity costs would not affect public provisions for the poor.

Our main hypothesis is that economic inequality reduces public goods provision, but a contested election with equal political strengths can mitigate or even eliminate the adverse effects. Hence, an incumbent who is either strong or weak is less likely to provide for the poor.

To test the hypothesis, we estimate how political competition and income inequality impact basic healthcare provision and MGNREGA employment. To quantify the effect on health outcomes, we also estimate the reduced form impact on post-neonatal infant mortality. The impact of inequality and political competition is interesting per se, but our main focus is on their interaction: Can higher political competition mitigate the negative impact of economic inequality?

III Institutional Background

Here we provide the context for the mechanism we test in this study. We describe the role of members of parliament in local public good provision, and the electoral redistricting of 2008.

III.A Members of Parliament and Public Good Provision

India has four administrative levels (states, districts, sub-districts and villages/wards), and first-past-the-post elections at five levels (lower house of the parliament, state assembly, district council, sub-district council and village council). One member of parliament (MP) is elected in each of the 543 parliamentary constituencies (PCs). The constituencies do not cross state borders, but may cross the boundaries of administrative districts. The three bottom levels make up the so-called Panchayati Raj system of local governance. Our main analysis focuses on the parliamentary level but – as we outline in Section IV – we systematically control for district and state-level differences by including district fixed effects, and thus implicitly, also state fixed effects. Below, we briefly discuss public healthcare, the MGNREGA public work program and the role MPs play in local public good provision.

Public healthcare. In rural India, the government healthcare system consists of three tiers. The lowest tier is the sub-centers, which are supposed to cover a population of 3,000 to 5,000 and have a sanctioned strength of one male and one female health worker. Primary

Health Centers (PHCs) make up the second tier, and cover a population of 20,000 to 30,000. They provide curative, preventive and promotive healthcare, and act as a referral unit for the sub-centers. The third tier, the Community Health Centers, provide specialist care and act as referral centers for the PHCs. In this paper we focus on the PHCs, which are the cornerstone of the rural health system in India and the first contact point between villagers and government medical officers.

The state governments have the main responsibility for the provision of public health- care, but the district and parliamentary level also play a crucial role. Officials at the district level gather data on healthcare demands from local governments, and based on this, present budget proposals to the state governments. The allocation of funds across local governments is mostly decided at the district level as well (Kailthya and Kambhampati 2016). Health is also a recurrent topic in the Parliament of India. Indeed, from the digital library, we learn that questions from MPs to the ruling government have *health* as a subject as often as *employment* and *unemployment* combined, and about four times as often as *drinking water*.[8]

MGNREGA. Guaranteeing 100 days of minimum-wage employment per year to each rural household, MGNREGA is the world's largest employment scheme. It is funded jointly through central and state government budgets, but is implemented at the local level. Typical work consists of building and maintaining local public goods such as wells, ponds and dams. Its implementation is highly uneven across India. Gulzar and Pasquale (2017) show that the top decile of households in 2013 worked 98 days per year while the bottom decile worked only 17 days. They also document large variation even within small geographical areas. In principle, this variation could be due to differences in the demand for work, but earlier research has demonstrated that the differences are almost entirely due to the supply side (Dasgupta and Kapur 2020; Dutta et al. 2014; Khosla 2011; Maiorano 2014).

MPs' local influence. MPs play a prominent role in local politics. They receive a yearly budget to implement development works in their constituencies, through the so- called Members of Parliament Local Area Development Scheme (MPLADS). The size of this program was 50 million rupees (about US$ 750,000) per MP per year in 2016, a 100-fold increase since it was initiated in 1993 (MoSPI, 2016). Several of the MPLADS projects are healthcare projects, such as the purchase of ambulances and equipment for local health centers (Swaminathan et al. 2019).

The MPs are also ex officio members of the district-level councils of all the districts that geographically overlap with their constituencies. The MPs are invited to local district council meetings, where the distribution of resources is discussed – including healthcare and MGNREGA. This gives them the opportunity to influence the distribution of local funds.

In addition, the MPs affect outcomes at the local levels through informal channels, for instance by influencing and pressuring local bureaucrats. Gupta and Mukhopadhyay (2016) use data from Rajasthan to show that larger amounts of MGNREGA funds were allocated to blocks where the incumbent party at the state level – the Indian National Congress (INC) – had a lower seat share. This effect is only found in districts where the MP was from INC, suggesting that MPs are indeed able to affect the implementation of the program. Moreover, Maheshwari (1976) showed that MPs, in the 1970s, spent considerable time handling requests from their constituents. Most of these requests fell within the jurisdiction of the state and not the central government. More recently, Bussel (2018) finds that MPs in Bihar, Jharkhand and Uttar Pradesh spend close to a quarter of their reported working time meeting their constituents.

III.B The 2008 Delimitation

When the Constitution of India was drafted, the plan was to update the electoral boundaries every ten years to (i) equalize population sizes across constituencies and (ii) reserve constituencies for Scheduled Castes (SCs) and Scheduled Tribes (STs) in proportion to up-to-date measures of their population shares. In the 1970s, however, it was decided to keep the political boundaries fixed for three decades because the increasing political representation of areas with a higher birthrate created a counter-incentive to the implementation of family planning programs (Jensenius 2013).

The process of redrawing the map started in mid-2004 and was based on population characteristics from the 2001 Census. At the outset it was decided that the number of seats in the national legislature would remain fixed for each state. Thus, the redistricting exercise only reallocated voters across constituencies within Indian states. The redistricting was organized by an independent three-member Commission. In each of the states, this Commission was advised by ten associate members, consisting of five MPs and five members of the State Assembly (MLAs). These associate members had no formal voting power, but they were closely involved in the process. A draft of the new electoral boundaries was distributed widely and public comments were invited. The Commission submitted a final report which was approved by the President of India in August 2008. In the end, about one quarter of the rural population was shifted to a new constituency (Table B1 in Online Appendix B provides this statistic by state).

Many countries go through similar processes and influential politicians are often accused of tweaking the redistricting to create safer seats for themselves. Although the Delimitation Commission was independent, we cannot be absolutely certain that the politicians did not have an influence on the outcomes. Iyer and Reddy (2013) study the redistricting of State Assembly constituencies in two large states, Andhra Pradesh and Rajasthan, and conclude that the boundary changes were politically neutral for most parts. Bardhan et al. (2020) study this redistricting in West Bengal and come to the same conclusion.

To validate our identification, we conduct a similar analysis as Iyer and Reddy

(2013) for PCs in the 15 states in our sample.[9] We present the details of this exercise in On- line Appendix B, but we summarize the main results here. First, we confirm that the redistricting was done in the intended direction: The purpose was to equalize population sizes within states and the greatest absolute population changes occur in the smallest and largest constituencies, which is consistent with this goal. Second, we do not find evidence of gerrymandering. We derive this conclusion by defining a set of influential MPs who could have been in a position to influence the process and investigating whether these politicians experienced an ex-ante more favorable redistricting than others. We do not find evidence of this.

These results are based on variation across constituencies and do not serve as a direct test of our identification, which is based on variation within constituencies (see the next section). Still, they suggest that demographical rather than political motives drive the boundary changes, and so it is appropriate to use the Delimitation as a source of exogenous variation.

IV Empirical Setup

In this section, we first explain how we exploit the boundary changes to identify the causal impact of income inequality and political competition on our outcome variables of interest. We then detail the data sources, discuss the sample selection, and assess the empirical model.

IV.A Identification

Figure 1 illustrates our identification. Panel I illustrates the boundaries of two pre-delimitation constituencies *A* and *B*, and Panel II illustrates the boundaries of the corresponding post-delimitation constituencies A^I and B^I. Consider two locations, marked by *a* and *b* in the figure. Before the Delimitation they belonged to the same constituency *A*. As a consequence of the boundary changes, however, location *b* is "dropped" into constituency B^I and location a into constituency A^I. Our approach is to compare locations such as a and b, which – due to the Delimitation – no longer face the same levels of inequality and political competition in their constituency.

Households that are allocated to a new constituency may directly influence the explanatory variables of interest in their new constituency. For example, if poor families are shifted to a relatively richer constituency, inequality in the new constituency would mechanically be larger. Poor households are more likely than rich households to have higher mortality and demand different types of social support. To deal with this, and to avoid capturing spurious associations between inequality and our outcomes of interest, we calculate the constituency-level variables using the boundaries of the pre-delimitation constituencies.

We define the corresponding pre-delimitation constituency as the one that has the largest population overlap with the post-delimitation constituency of interest. This implies that each post-delimitation constituency is linked to one unique pre-delimitation constituency. In terms of Figure 1, this means that we use the characteristics of pre-delimitation constituency A for location a and the characteristics of pre-delimitation constituency B for location b. For locations that changed constituency due to the Delimitation – such as b – this means that the location itself does not enter the calculation of the main explanatory variables.

Our regressions include fixed effects for pre-delimitation constituencies interacted with districts (the most important administrative unit for social provision to the poor).[10] Our strategy is thus to compare locations that were allocated to a new constituency (for instance, location b) with locations in the same district and the same pre-delimitation constituency, but that were not allocated to the new constituency (for example, location a). In the mortality regressions we also interact the fixed effects with the birth year of infants to account for possible time trends in mortality, and in the MGNREGA regressions we interact the fixed effects with the fiscal year to account for possible time trends in its provision.

The key identification assumption underlying our analysis is that the relative supply of healthcare and MGNREGA, and the relative mortality risk in locations such as *a* and *b*, are only affected by the changes in inequality and political competition at the constituency level. For this to be the case, the allocation of households to constituencies within a district should be as-good-as-random. Given the fine geographical level of our identifying variation and how the Delimitation was implemented, we think this is a plausible assumption. We conduct two tests to examine it. First, we look for possible differences in observables between villagers who were allocated to new constituencies and those who remained in their original constituencies. Second, we run a placebo test using information on health centers and infants in locations *a* and *b* *before* the Delimitation. As the locations belonged to the same pre-delimitation constituency at that time, their relative healthcare provision and mortality risk should not be affected by differences in inequality and political competition in the constituencies they are shifted to after the Delimitation.[11]

The final concern is that inequality and political competition may pick up the effects of other correlated changes at the constituency level. We cannot entirely exclude this possibility, but we believe it is unlikely. First, we identify our main effects using characteristics of neighbouring constituencies. For example, other studies document a positive relationship between income inequality and infant mortality, which reflects – at least partially – the concentration of infant mortality at the bottom end of the income distribution (Deaton, 2003). We exclude such a "direct" effect of inequality on mortality by calculating inequality based on neighbouring constituencies. Second, parliamentary constituencies do not *implement* local policies. As such, differences in their administrative and financial capacity should, at least in principle, have no explanatory power. These differences may matter at the level of districts, but we control for those through the district fixed effects. The key remaining factors that could threaten the empirical setup are, therefore, characteristics of the constituencies that impact the decisions taken by the MP. To alleviate some of this concern, we run a set of horse race regressions, where we interact our main variables of interest with a large set of constituency characteristics.

IV.B Specifications

We use the following specification to estimate the effect of inequality and political competition on mortality:

$$(1)\quad m_{idklt} = \beta_0 + \beta_1 v_l + \beta_2 z_l + \beta_3 (v_l \times z_l) + \beta_4 q_l + \theta'_{dkt} + X'_{1,dkl}\gamma_1 + X'_{2,idkl}\gamma_2 + \varepsilon_{idklt}$$

where m_{idklt} is the mortality of infant i born in year t in district d and pre-delimitation constituency k. The subscript l denotes which pre-constituency infant i's post-constituency is linked to. Our key variables are measured at the constituency level. As we do not have information on election outcomes below constituency level, we construct the measures of political competition, z_l, based on the election prior to the Delimitation. This election took place in April-May 2004 and should not have been affected by the Delimitation as the Commission only started its work in July 2004. Our approach thus presumes that political competition is persistent from one period to the next, which is supported by the data.[12]

Inequality is denoted by v_l and mean income by q_l. To ease the interpretation, we standardize these right-hand side variables by subtracting their mean and dividing by their standard deviation. The fixed effects are denoted by θ'_{dkt} and include the infant's birth year t to account for possible time trends in mortality. The standard errors, ε_{idklt}, are clustered at the district × pre-delimitation constituency level. We add two vectors of controls. First, we control for area characteristics at the pre-delimitation constituency× post-delimitation constituency × district level, denoted by $X'_{1,dkl}$. This includes the literacy rate, the population share of SCs and STs, the population share of children below six years of age and the availability of different public amenities prior to the Delimitation. Second, we also include a vector of child level characteristics, denoted by $X'_{2,idkl}$, which includes observables that may impact the likelihood of mortality: A binary variable denoting females, a binary variable denoting twins, and binary variables denoting religion (Hindu, Muslim, Christian, Sikh and Buddhist).

For the provision of public healthcare, we estimate:

$$(2)\quad S_{jdkl} = \beta_0 + \beta_1 v_l + \beta_2 z_l + \beta_3 (v_l \times z_l) + \beta_4 q_l + \theta'_{dk} + X'_{1,dkl}\gamma_1 + \varepsilon_{jdkl}$$

where S_{jdkl} captures the staff and service index for PHC j. Compared to Equation 1, we make two straightforward adjustments: There is no time trend to control for in the fixed effects θ'_{dk}, and we only control for the area characteristics $X'_{1,dkl}$.

Finally, we estimate the following specification for MGNREGA performance:

$$(3)\quad y_{jdklt} = \beta_0 + \beta_1 v_l + \beta_2 z_l + \beta_3 (v_l \times z_l) + \beta_4 q_l + \theta'_{dkt} + X'_{1,dkl}\gamma_1 + \varepsilon_{jdklt}$$

where y_{jdklt} denotes the MGNREGA performance in Gram Panchayat (GP) j in fiscal year t. The inclusion of the fiscal year t in the fixed effects θ'_{dkt} allows us to account for possible time trends in MGNREGA provision.

As stated in Section II, we are mainly interested in the coefficient β_3, capturing the interaction effect. In the mortality regressions we expect $\beta_3 < 0$ and $\beta_1 > 0$, according to our hypothesis, as political competition should dampen the "harmful" impact of inequality. For the same reason we expect $\beta_3 > 0$ and $\beta_1 < 0$ in the public healthcare and MGNREGA regressions, as higher values of the left-hand side variable are associated with better outcomes in this case.

IV.C Data

In this section, we describe our data sources and main variables. Table 1 summarizes the sources we link through geocodes, starting from the ML InfoMap village and constituency maps. We provide a more detailed description of the data construction and merging in Online Appendix E.

Mortality. We use the 2015-2016 National Family and Health Survey (NFHS) as our data source on mortality.[13] The NFHS interviews women aged 15-49 and measures their complete birth record. The survey data contain information on the timing of all births, and if the child died, the age in months when death occurred. This allows us to construct a retrospective time series of infant deaths.[14] The 2015-2016 survey interviewed about 700,000 women, which is a much larger sample than in the previous NFHS waves, facilitating the study of mortality at a fine geographical level. Another advantage is that the 2015-2016 survey provides GPS coordinates for survey clusters. These clusters roughly coincide with the GPs. We combine the cluster coordinates with the geocoded maps of constituency boundaries to allocate survey respondents to constituencies.

We distinguish between neonatal and post-neonatal infant mortality. Neonatal mortality is defined as deaths before four weeks of age, but – as in Bhalotra and van Soest (2008) – we include deaths up to one month to allow for age-heaping. Post-neonatal infant mortality is then defined as deaths between one and twelve months of age. We distinguish between the two types of mortality as the practices needed to reduce them are different. Although post-neonatal infant deaths can be reduced through interventions that focus on pneumonia, diarrhea and vaccine-preventable conditions, achieving major reductions in early neonatal deaths will depend on the provision of individualised clinical care (Lawn, Cousens and Zupan 2005). Therefore, we focus on post-neonatal infant mortality in the main results and discuss neonatal mortality in Online Appendix D.3 only.

For both neonatal and post-neonatal infant mortality, we construct a binary variable taking the value of one if the child died and zero otherwise, provided that the child was fully exposed to the particular mortality concept (see for example Rutstein 2005). To clarify, this means that the sample used to calculate post-neonatal infant mortality only includes children born at least 12 months before the end of the period, whereas the sample for neonatal mortality includes children born at least one month before the end of the period.

Public health centers. We obtain data on PHCs from the District Level Household and Facility Survey (DLHS) for 2007-2008 and 2012-2013. The 2012-2013 DLHS wave provides geocodes for the location of the surveyed PHCs and this enables us to link the data to the Census village map. The 2007-2008 wave does not contain such geocodes. We therefore follow a procedure similar to Banerjee and Sachdeva (2015) and Calvi and Mantovanelli (2018) and match villages to the Census based on state, district, sub-district and population. We are able to unambiguously match about 90% of the DLHS villages in this way (see Online Appendix E.3 for more details).

Both DLHS survey waves provide information on a large number of PHC characteristics. Given the multiplicity of hypotheses to test, we construct one main index for each survey wave, capturing the availability of key medical staff and the services they provide. Regarding medical staff, we constructed three binary variables denoting whether the PHC has a doctor, a nurse/midwife and a health visitor. The latter started operating in 2005 as part of India's National Rural Health Mission. Björkman et al. (2019) show that the introduction of community health workers reduced child mortality in Uganda. We also constructed binary variables denoting whether the PHC provides medical services that are particularly relevant for reducing child mortality: whether the PHC (i) provides antenatal care; (ii) conducts deliveries; (iii) has management services available for diarrhea and pneumonia; (iv) provides vaccinations, and (v) conducts health check-ups in schools.

We combine these variables into a summary index, using the procedure from Anderson (2008). We first convert each variable into a z-score, by subtracting the sample mean and dividing by the standard deviation. We then generate the summary index by averaging the z-scores, using weights generated by the inverse of the covariance matrix. In this way, the procedure gives more weight to uncorrelated information. Finally, we standardize the overall index to ease interpretation, again by subtracting the sample mean and dividing by the standard deviation.

MGNREGA. We extract GP level data from the MGNREGA Public Data Portal for the financial years 2011-12, 2012-13 and 2013-14. This data includes names of districts, sub-districts and GPs but it has no information on Census identification numbers. We merge the data to the Census based on fuzzy matching of GP names (within states, districts and sub-districts). In total, we are able to match 76.5% of the 2011 Census villages to the MGNREGA dataset, which is comparable to the work of other researchers doing fuzzy matching in the Indian context (Asher and Novosad 2017; Gulzar and Pasquale 2017). We provide more details on the data compilation in Online Appendix E.4. Using this data, we construct two outcome variables: (i) the total amount disbursed to laborers' bank and post office accounts, and (ii) the number of households demanding work. We use the latter variable as a placebo outcome.

Electoral outcomes. Data on electoral outcomes are taken from the Indian National Election and Candidates Database (Jensenius and Verniers 2017). This dataset contains the number of votes the most important candidates received in each constituency. We use the election results from 1999 and 2004. Consistent with the theoretical motivation in Section II, we measure political competition using the vote shares of the incumbent and their opponents as one minus the Herfindahl-Hirschman index (HHI).

In the theoretical model in Online Appendix A, we derived this by representing the opposition as one group. A natural extension to this is to use the detailed information on vote shares for each of the opposition candidates (see, for instance, Callen et al. (2023) for a recent application of this measure of political competition in Pakistan). A high value of the Herfindahl-Hirschman index implies that the votes cast have a high concentration on one candidate (either the incumbent or one of the challengers), meaning that the electoral pressure is likely to be low. For constituency *l*, the political competition is measured by:

$$1 - HHI_l = 1 - \sum_{c=1}^{n} s_{cl}^2$$

where s_{cl} is the vote share of candidate c. There are usually a large number of candidates running for elections in India. Most of these candidates attract a trivial vote share, and hence do not influence the above measure much. We limit the Herfindahl-Hirschman index to the ten most important candidates in each constituency.[15] In Section V.D, we investigate whether our results are robust to other measures of political competition.

Average expenditures and inequality. Data on household expenditures, which we use to calculate average expenditures and inequality, are taken from the National Sample Survey (NSS). The Gini coefficient is our baseline measure of economic inequality, but we also investigate whether our results are robust to other measures.

The NSS expenditure survey is a national-wide representative household survey, usually conducted every five years, with a sample size of more than 100,000 households in each round. We use the 2009-2010 survey (66th round) for our main analysis and the 2004-2005 survey (61st round) for our placebo analysis.

Our analysis requires estimates of mean expenditures and inequality at the level of election constituencies. However, the NSS data does not include identifiers for constituencies, nor does it provide geocodes for where households are located. The finest geographical unit we can identify is the district. Sometimes these districts perfectly coincide with constituencies, sometimes they do not. Another challenge is that some district boundaries changed during our study period. We tackle this as follows. We first convert the districts that changed between 2001 and the NSS surveys back to the district boundaries as of the 2001 Census. In most cases a single district was split into two parts, so this adjustment is straightforward. Based on the geocoded maps of election constituencies and 2001 Census villages, we then calculate the population share of each district in the different constituencies. A share can be interpreted as the probability that a household belongs to a particular constituency, conditional on the district it resides in.[16] Finally, we calculate mean expenditures and inequality for a constituency by weighting households by their probability of residing in that particular constituency.[17]

Area controls. We construct a set of control variables using the Census of India for 2001. This dataset includes basic population characteristics and information about a large number of amenities for all Indian villages.

IV.D Sample Selection and Summary Statistics

The Delimitation changed electoral boundaries within states, while the number of constituencies for each state remained the same. Because the small states of India only have one or two constituencies each, we exclude them from the analysis. We also exclude Assam, Jammu & Kashmir and Jharkhand because they did not implement the Delimitation. We are left with a sample of 15 states, which are listed in Online Appendix Table B1. According to the 2011 Census, these states account for about 90% of the total rural population.

We limit our main analysis to rural areas of India.[18] There are two reasons for doing so. First, our identification assumption may not hold for urban areas, as it is much less likely that cities are allocated to new constituencies than rural areas. Second, most urban areas have access to a wide variety of publicly provided goods and services, making our proposed mechanism less relevant. Note, however, that political competition, average expenditures and inequality are measured at the constituency level, which may include urban areas.

In our mortality regressions we further limit the sample to infants born (and fully exposed to the particular mortality concept) in the time period between the 2009 and 2014 parliamentary elections. We exclude infants whose mothers moved after their birth, as we do not know where these infants spent the first year of their lives. The same applies for infants whose mother was a visitor in the location where she was interviewed.[19]

Table 2 provides summary statistics for the main variables. About 1.4% of children die after the first month but before age one. As the healthcare index is standardized, it has mean zero and standard deviation one. The average GP disburses 775,559 rupees (US$ 12,118 in 2015) per fiscal year to laborers' bank and post office accounts. The average Gini coefficient across constituencies is 0.28 and our measure of political competition, one minus the Herfindahl-Hirschman index, equals 0.62.

IV.E Assessing the Empirical Model

Our identification assumes that changes in exposure to constituency-level inequality and political competition are unrelated to other factors affecting our outcomes once we include our fixed effects. To investigate whether our data supports this claim, we implement two balance tests. Table 3 and Online Appendix Table D1 present the results.

We first look for possible differences in observables between households, GPs and PHCs allocated to new constituencies and those that remain in their original constituencies. To do so, we run the following regressions:

$$X_{idkl} = \beta_0 + \beta_1 T_{idkl} + \theta'_{dkt} + \varepsilon_{idkl} \text{ for child controls (mortality sample)}$$

$$X_{dkl} = \beta_0 + \beta_1 T_{idkl} + \theta'_{dkt} + \varepsilon_{idkl} \text{ for area controls (mortality sample)}$$

$$X_{dkl} = \beta_0 + \beta_1 T_{jdkl} + \theta'_{dk} + \varepsilon_{dkl} \text{ for area controls (healthcare and MGNREGA sample)}$$

where i denotes a household, j a PHC or a GP, X_{idkl} a child-level control and X_{dkl} an area-level control. T is a binary variable indicating whether the household, GP or PHC was allocated to a new constituency, and θ defines the fixed effects.[20] Column 1 of Table 3 displays the coefficient of interest, β_1, for each of the area and child-level controls using the mortality sample. Columns 2 and 3 show the concurrent estimates for the area controls using the healthcare and MGNREGA samples, respectively. Locations that changed constituency do not differ significantly from those that did not.[21] The last line provides the share of the redistricted observations in each sample (about 26%).

As a second balance test, which we present in Online Appendix Table D1, we regress our main specifications, using the controls as dependent variables. Overall, the results suggest that our key constituency-level variables of interest are unable to predict area and individual characteristics. There is some imbalance for religion only.

V Results

In this section, we first estimate the impact of income inequality and political competition on our outcomes of interest. As predicted by our theoretical discussion, we find that higher economic inequality leads to more post-neonatal infant deaths and worse performance of public healthcare and MGNREGA, but only in situations with weak political competition. We then underline the robustness of these results, including a placebo analysis and horse race regressions, among other checks.

V.A Main Results

Table 4 provides the main results for post-neonatal infant mortality (Columns 1-2), the healthcare index (Columns 3-4) and the total amount disbursed to MGNREGA workers' bank and post office accounts (Columns 5-6). We control for average expenditures in all columns and in the even columns for the area characteristics. In Column 2, we also include child-level variables. Given the seemingly balanced sample, it is not surprising that the results in the even columns closely match those in the uneven columns.

Mortality. We first estimate the effects of inequality and political competition on post-neonatal infant mortality using specification 1. At average levels of political competition, inequality does not significantly impact mortality. This can be seen directly from the inequality coefficient (the independent variables are standardized). However, the interaction term is negative and significant. To interpret the magnitude of this effect, suppose that the Gini coefficient increases by one standard deviation (corresponding to a

6.2 percentage points increase). If political competition is one standard deviation below its sample mean, this rise in inequality leads to an increase in post-neonatal infant mortality of 0.18 percentage points, or 13% of the sample mean, which is substantial.[22] Online Appendix Table D2 presents the estimates for neonatal mortality.

Healthcare provision. The results on public healthcare provision are based on specification 2. The outcome variable of interest is the healthcare index, which reflects the key staff and services provided at PHCs. As for post-neonatal infant mortality, we find no impact of inequality at average levels of political competition. However, the interaction term is positive and significant. Suppose, as above, that the Gini coefficient increases by one standard deviation. If political competition is one standard deviation below its sample mean, this rise in inequality leads to a decrease in the healthcare index of 0.15 standard deviations.[23]

In Online Appendix Table D4, we also look at the demand for public healthcare using a survey question asking about the type of health centers that households "usually go to in the case of illness". The results show that higher economic inequality, combined with weak political competition, makes it significantly less likely that households answer that they go to a government health center. This finding suggests that families respond to the weaker functionality of PHCs in these areas.

MGNREGA. In the final two columns of Table 4, the outcome variable is the total amount disbursed to laborers' bank and post office accounts per year and GP. For ease of interpretation, we standardize it to mean zero and standard deviation one. We find a similar pattern as for public healthcare: When political competition is at its sample mean, a one standard deviation rise in inequality reduces the amount disbursed by about 0.018 standard deviations. If political competition is one standard deviation below the mean, the same rise in inequality reduces the amount disbursed by 0.028 standard deviations, which corresponds to 48,500 rupees (6.2% of the sample mean) per year.

In conclusion, consistent with our theoretical discussion, economic inequality leads to i) more post-neonatal infant deaths, ii) worse functioning public health clinics and iii) less public work – but only in situations with weak political competition. We have identified these effects using three independent datasets. While the results on PHCs and MGNREGA are closely linked to our proposed theoretical mechanism, we interpret the mortality results as a reduced-form effect. We do not claim, however, that the mortality results are driven exclusively by a worsening of health clinics and MGNREGA employment. It is plausible that healthcare is an important driver, as access to basic health products and services has been shown to reduce infant mortality in other settings (Baqui et al. 2008; Björkman et al. 2019; Kumar et al. 2008). MGNREGA could also play a role, as it has raised wages (Imbert and Papp 2015) and improved living conditions more generally (Ministry of Rural Development 2012). The link with child health and survival is, however, somewhat unclear (Chari et al. 2019; Spears and Lamba 2016). On the one hand, the program emphasises female participation, which could improve female empowerment and investment in children. On the other hand, female participation might crowd out time spent by mothers with their children, which is likely to reduce child health.

Other types of social provision, such as clean drinking water and sanitation, could also impact the mortality results (and are equally consistent with our proposed mechanism). We also cannot exclude the possibility that other economic factors are affected by changes in inequality and political competition, and are driving the mortality results. In Online Appendix D.5, we test the latter hypothesis for the village-level economic outcomes we have information on: employment per capita, number of firms per capita and the average amount of night-time light.[24] We do not find a significant interaction between inequality and political competition for these outcomes.

In the remainder of this section, we explore the robustness of our findings by conducting a placebo analysis, running horse race regressions and testing several other alternative specifications.

V.B Placebo Regressions

Table 5 presents placebo regressions. For mortality and healthcare, we use data from *before* the Delimitation. In particular, we utilise the NSS survey from 2004-2005 to calculate mean expenditures and inequality and outcomes from the 1999 parliamentary election to calculate political competition.

For post-neonatal infant mortality, we restrict the NFHS sample to infants born between the 2004 and 2009 parliamentary elections. These children should not have been affected by electoral outcomes in their (future) post-delimitation constituencies. The estimates in Column 1 suggest they indeed were not. The interaction term is negative, but the coefficient is not statistically significant, and its magnitude is about one-third of the concurrent coefficient in our main regression (Column 2 in Table 4).

For healthcare provision, we construct the same healthcare index, but using data from the 2007-2008 DLHS. The results in Column 2 show that none of the estimated coefficients are statistically significant. The interaction term has the same sign as before, but the magnitude is much smaller.[25] We also run a placebo regression for the demand for public healthcare. Column 2 in Online Appendix Table D4 shows that none of the estimated coefficients are statistically significant.

We cannot run a similar placebo test for MGNREGA, as disaggregated data is avail- able only from 2011-2012 onwards. Instead, we use data on the number of households demanding MGNREGA work. As mentioned before, research has shown that differences in implementation are almost entirely due to the supply side. Column 3 confirms that the observed differences in MGNREGA disbursement (Column 6 in Table 4) are not driven by the demand for work: The coefficients are much smaller than those in the main table and they are not significantly different from zero.

V.C Horse Race Regressions

Our interaction term of interest may pick up other factors correlated with inequality and/or political competition. To remove some of this concern, we run a set of horse race regressions where we interact inequality and political competition separately with other constituency-level characteristics. Table 6 shows the estimates for post-neonatal infant mortality in Panel A, the healthcare index in Panel B, and MGNREGA disbursements in Panel C. In Columns 1 to 7, we add interactions with one variable at a time: mean expenditure, poverty rate,[26] urbanization rate, total population, caste fragmentation, religious fragmentation,[27] and literacy rate. In Column 8, we present the results when we include all these variables at once. Finally, Column 9 includes interactions with all our main specifications' control variables. To ease the interpretation, we standardized the constituency-level variables.

Our main coefficient of interest is robust to most of these rather demanding specifications. In Panel A, the interaction coefficient is less precisely estimated in Column 8 but quantitatively similar. We draw a similar conclusion for the coefficients in Panels B and C, except for Column 9 in Panel C: In the most demanding specification, the coefficient is smaller in magnitude (compared to Column 6 in Table 4). With few exceptions, the interactions with the other constituency-level variables are not significantly different from zero at a 10%-level.[28]

V.D Other Robustness Tests

We present further robustness tests in Table 7. In Columns 1-3, we explore whether our results are robust to using the following alternative measures of inequality: (i) the ratio of the average income of the wealthiest 10% to the poorest 40%, also known as the Palma ratio, (ii) the mean log deviation,[29] and (iii) the Theil index. Overall, the results are stronger than our baseline estimates, providing additional evidence in support of our main hypothesis. Indeed, our mechanism relies on important income gaps between the rich and the poor. The Gini coefficient is most sensitive to inequalities in the middle of the income spectrum, and the alternative measures are more sensitive to inequalities between top and bottom incomes.

Columns 4-6 in Table 7 explore the robustness of our results to alternative measures of political competition. In particular, we investigate the following measures: (i) one minus the vote share of the winning party, (ii) one minus the margin of victory (MoV), and (iii) the effective number of parties (ENOP). For each of these measures, a higher value indicates more political competition. The MoV is defined by the difference in vote shares between the winner and the runner-up, while the ENOP, developed by Laakso and Taagepera (1979), is measured as:

$$ENOP_l = \frac{1}{\sum_{c=1}^{n} s_{cl}^2}$$

where s_{cl} is the vote share of candidate c in PC l.[30] The measure runs from one (if one candidate got all the votes) to the maximum number of candidates that ran in the election (if each got the same vote share). Our results are robust, except for the interaction coefficient for (1-MoV), which is less precisely estimated. This is not entirely surprising, as this measure is commonly used for two-party settings such as the United States (Cox, Fiva and Smith 2020), but is not as well suited capturing political competition in settings with more than two relevant political parties (see Callen et al., 2023).[31]

We report additional robustness tests in the appendix. In Online Appendix C.2, we added dummy variables for the five most prominent political parties in 2009 to check whether our findings are driven by party ideology or the party that rules nationally rather than by political competition (Asher and Novosad 2017). We also added controls for the reservation status of the constituencies in 2009. In addition, we present the results for both these tests after including an interaction of inequality and political competition with the party dummies and the reservation status, respectively. Our estimates are mostly robust to these specifications.[32] In Online Appendix C.3, we restrict our sample to observations with a maximum distance of 40 or 20 kilometres to their closest post-delimitation constituency border. We lose some precision, but the results are otherwise robust to this sample selection.

Finally, politicians and bureaucrats at lower levels, such as blocks and assembly constituencies (ACs), may also influence local public goods provision. There can be substantial variation across blocks and ACs, but – given that these map entirely into one district – the concern is whether there is balance, that is whether areas allocated to new constituencies and those that remain in their original constituencies are balanced on relevant block and AC characteristics. Online Appendix C.4 shows the sample imbalances are minor.

VI Conclusion

Our paper focuses on the roles of political competition and economic inequality at the local level. We show that an incumbent with a high probability of winning, or a high probability of losing, is tempted to cater more to the wealthy than to the poor if the income inequality is high. In such situations, the social provision for the poor may be so low that it threatens children's health. Elections that are sufficiently contested, however, can discipline the incumbent, as it makes it too politically costly to ignore social provision to the poor even when inequality is high.

We find empirical support for this hypothesis in the case of India. We use a large redistricting of electoral boundaries to obtain exogenous variation in measures of political competition and income inequality and explore their reduced form impact on infant mortality and measures of social provision that can substantially reduce its prevalence. Higher economic inequality leads to more post-neonatal infant deaths, but only in situations with weak political competition. We argue that the effect on mortality operates via changes in public provision at the local level. Indeed, government health centers in constituencies with low political competition and high inequality are worse off: They have fewer staff and provide fewer services. These constituencies also see a weaker provision of MGNREGA, a program that aims at improving social protection for the most vulnerable.

In conclusion, our study makes a case for the importance of democratic accountability in the context of a middle-income country. Without sufficient electoral pressure, politicians in areas with high inequality do not offer enough social provision for the poor, and the consequences can be fatal.

Tables

Table 1: Overview of data sources

Variable	Source	Main analysis	Placebo analysis
Mortality	National Family and Health Survey (NFHS)	2015-2016	2015-2016
Public healthcare	District Level Household and Facility Survey (DLHS)	2012-2013	2007-2008
MGNREGA	MGNREGA Public Data Portal	2011-2014	2011-2014
Electoral outcomes	Indian National Election and Candidate Database	2004	1999
Average expenditures and inequality	National Sample Survey	2009-2010	2004-2005
Area controls	Census of India	2001	2001

Table 2: Summary statistics

	Level (1)	Observations (2)	Mean (3)	Std. Dev. (4)
Post-neonatal infant mortality	Individual	90,236	0.014	0.115
Public healthcare index	PHCs	5,679	0.000	1.000
MGNREGA	GPs	451,231	775,559	1,734,302
Gini coefficient	Constituency	447	0.276	0.062
1-Herfindahl-Hirschman of vote shares	Constituency	447	0.617	0.081
Mean expenditures	Constituency	447	1,124	393

Table 3: Balance table

	Mortality sample (1)	Healthcare sample (2)	MGNREGA sample (3)
Panel A: Area characteristics:			
Share of Scheduled Castes	0.0004	-0.0000	0.0006
	(0.0033)	(0.0038)	(0.0031)
Share of Scheduled Tribes	0.0032	0.0139*	0.0096
	(0.0054)	(0.0082)	(0.0076)
Share of children below six years old	-0.0001	-0.0002	-0.0006
	(0.0008)	(0.0008)	(0.0007)
Share of literate people	-0.0037	-0.0064*	-0.0065*
	(0.0035)	(0.0038)	(0.0035)
Share of villages with a primary school	-0.0019	0.0013	-0.0030
	(0.0074)	(0.0056)	(0.0049)
Share of villages with a PHC	0.0023	-0.0001	-0.0010
	(0.0019)	(0.0018)	(0.0017)
Share of villages with a PHC sub-centre	0.0011	0.0051	0.0040
	(0.0049)	(0.0066)	(0.0043)
Share of villages with tap water	-0.0038	-0.0052	-0.0006
	(0.0097)	(0.0100)	(0.0108)
Share of villages with electricity	-0.0063	-0.0108	-0.0100
	(0.0104)	(0.0108)	(0.0083)
Share of villages with paved road	-0.0054	-0.0136	-0.0052
	(0.0079)	(0.0092)	(0.0075)
Panel B: Individual characteristics:			
Child is a girl	-0.0065		
	(0.0057)		
Child is a twin	0.0004		
	(0.0022)		
Religion: Hindu	0.0107		
	(0.0111)		
Religion: Muslim	-0.0075		
	(0.0106)		
Religion: Christian	-0.0004		
	(0.0015)		
Religion: Sikh	-0.0021		
	(0.0035)		
Religion: Buddhist	-0.0020*		
	(0.0012)		
Observations	90,236	5,679	150,413
Share redistricted	0.2666	0.2622	0.2692

Column 1 reports the differences between villagers who were allocated to new constituencies and those who were not for the mortality sample. Columns 2 and 3 do the same exercise for PHCs and GPs in the healthcare and mortality samples, respectively. Robust standard errors, clustered at the pre-delimitation constituency × district level, are shown in parentheses. * significant at 10 percent.

Table 4: Main results

	Post-neonatal infant mortality		Healthcare index		NREGA disbursements	
	(1)	(2)	(3)	(4)	(5)	(6)
Inequality	0.0002	-0.0001	-0.0600	-0.0672	-0.0115	-0.0178*
	(0.0016)	(0.0016)	(0.0561)	(0.0574)	(0.0113)	(0.0099)
Political competition	0.0003	0.0007	0.0181	0.0225	-0.0016	-0.0015
	(0.0015)	(0.0016)	(0.0531)	(0.0530)	(0.0093)	(0.0093)
Inequality ×	-0.0018**	-0.0019**	0.0807**	0.0851**	0.0165**	0.0104*
Political competition	(0.0009)	(0.0009)	(0.0360)	(0.0372)	(0.0066)	(0.0061)
Observations	90,236	90,236	5,679	5,679	451,231	451,231
Average expenditures	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes
Unit	Infant	Infant	PHC	PHC	GPs	GPs

The independent variables are standardized, and the coefficients thus reflect the impact at average levels. The even columns control for average expenditures and area characteristics. In Column 2, we also control for child characteristics. Columns 1-2 include pre-delimitation constituency × district × birth year fixed effects, Columns 3-4 pre-delimitation constituency × district fixed effects, and Columns 5-6 pre-delimitation constituency × district × fiscal year fixed effects. Robust standard errors, clustered at the pre-delimitation constituency × district level, are shown in parentheses. ** significant at 5 percent, * significant at 10 percent.

Table 5: Placebo regressions

	Post-neonatal infant mortality (1)	Healthcare index (2)	MGNREGA demand (3)
Inequality	-0.0037	0.0604	-0.0006
	(0.0024)	(0.0560)	(0.0094)
Political competition	-0.0035	0.0312	0.0033
	(0.0024)	(0.0413)	(0.0105)
Inequality ×	-0.0007	-0.0165	0.0011
Political competition	(0.0016)	(0.0254)	(0.0051)
Observations	87,976	5,702	451,231
Average expenditures	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Unit	Infant	Clinic	GPs

The independent variables are standardized, and the coefficients thus reflect the impact at average levels. The regressions control for average expenditures and area characteristics. In Column 1, we also control for child characteristics. Column 1 includes pre-delimitation constituency × district × birth year fixed effects, Column 2 pre-delimitation constituency × district fixed effects, and Column 3 pre-delimitation constituency × district × fiscal year fixed effects. Robust standard errors, clustered at the pre-delimitation constituency × district level, are shown in parentheses.

Table 6: Horse race regressions

	Additional variable:								
	Mean expenditure (1)	Poverty rate (2)	Urbani-zation (3)	Population (4)	Caste frag. (5)	Religion frag. (6)	Literacy (7)	All (8)	Controls (9)
Panel A: Post-neonatal infant mortality (N=90,236):									
Inequality	0.0009	-0.0017	0.0000	-0.0001	-0.0005	0.0003	-0.0002	-0.0025	-0.0009
	(0.0017)	(0.0024)	(0.0016)	(0.0016)	(0.0016)	(0.0015)	(0.0017)	(0.0025)	(0.0043)
Political competition	0.0006	0.0006	0.0005	0.0009	0.0013	0.0011	0.0007	0.0015	0.0005
	(0.0016)	(0.0016)	(0.0016)	(0.0015)	(0.0016)	(0.0016)	(0.0016)	(0.0017)	(0.0050)
Inequality ×	-0.0024*	-0.0021**	-0.0025**	-0.0019**	-0.0022**	-0.0019**	-0.0022**	-0.0025	-0.0020*
Political competition	(0.0014)	(0.0010)	(0.0011)	(0.0008)	(0.0009)	(0.0008)	(0.0011)	(0.0018)	(0.0012)
Panel B: Healthcare index (N=5,679):									
Inequality	-0.0317	-0.0066	-0.0461	-0.0535	-0.0639	-0.0604	-0.0711	0.0498	-0.0549
	(0.0583)	(0.0852)	(0.0566)	(0.0573)	(0.0605)	(0.0562)	(0.0574)	(0.0907)	(0.0632)
Political competition	0.0168	0.0161	0.0108	0.0118	0.0383	0.0324	0.0049	0.0088	-0.0198
	(0.0515)	(0.0526)	(0.0513)	(0.0543)	(0.0556)	(0.0523)	(0.0513)	(0.0539)	(0.0528)
Inequality ×	0.0938*	0.0854**	0.0729*	0.0753*	0.0879**	0.0989**	0.0399	0.0993	0.0963*
Political competition	(0.0566)	(0.0401)	(0.0404)	(0.0409)	(0.0372)	(0.0388)	(0.0335)	(0.0645)	(0.0504)
Panel C: MGNREGA disbursement (N=451,231):									
Inequality	-0.0183*	-0.0274*	-0.0193*	-0.0183*	-0.0190*	-0.0161*	-0.0175*	-0.0150	-0.0204*
	(0.0107)	(0.0145)	(0.0108)	(0.0108)	(0.0100)	(0.0092)	(0.0097)	(0.0162)	(0.0115)
Political competition	0.0016	-0.0015	0.0025	0.0021	0.0009	0.0001	0.0005	0.0040	0.0015
	(0.0087)	(0.0093)	(0.0104)	(0.0109)	(0.0093)	(0.0089)	(0.0088)	(0.0111)	(0.0134)
Inequality ×	0.0182*	0.0124*	0.0154*	0.0134*	0.0120*	0.0128**	0.0153*	0.0282**	0.0013
Political competition	(0.0096)	(0.0065)	(0.0088)	(0.0072)	(0.0066)	(0.0062)	(0.0084)	(0.0133)	(0.0086)

The independent variables are standardized, and the coefficients thus reflect the impact at average levels. In Columns 1 to 7, inequality and political competition are interacted with one constituency-level variable at a time. The regression in Column 8 includes interactions with all the variables listed in Columns 1 to 7, and the regression in Column 9 includes interactions with all the area controls included in our main specification (and the interactions with the child characteristics in Panel A). The regressions control for average expenditures and area characteristics. In Panel A, we also control for child characteristics. Panel A includes pre-delimitation constituency × district × birth year fixed effects, Panel B pre-delimitation constituency × district fixed effects, and Panel C pre-delimitation constituency × district × fiscal year fixed effects. Robust standard errors, clustered at the pre-delimitation constituency × district level, are shown in parentheses. ** significant at 5 percent, * significant at 10 percent.

Table 7: Alternative measures of inequality and political competition

	Alternative measures of inequality			Alternative measures of political competition		
	Palma ratio	Mean log deviation	Theil	1-Win share	1-MoV	ENOP
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Post-neonatal infant mortality (N=90,236)						
Inequality	-0.0007	-0.0004	0.0004	-0.0005	-0.0009	-0.0002
	(0.0012)	(0.0014)	(0.0013)	(0.0016)	(0.0018)	(0.0015)
Political competition	0.0007	0.0008	0.0008	0.0007	-0.0000	0.0009
	(0.0016)	(0.0016)	(0.0016)	(0.0015)	(0.0015)	(0.0017)
Inequality ×	-0.0016**	-0.0020***	-0.0021***	-0.0017**	-0.0011	-0.0018*
Political competition	(0.0007)	(0.0007)	(0.0008)	(0.0008)	(0.0008)	(0.0010)
Panel B: Healthcare (N=5,679):						
Inequality	-0.1030**	-0.0912	-0.0787	-0.0561	-0.0348	-0.0694
	(0.0496)	(0.0563)	(0.0530)	(0.0554)	(0.0555)	(0.0573)
Political competition	0.0195	0.0248	0.0179	-0.0017	0.0084	0.0005
	(0.0514)	(0.0531)	(0.0525)	(0.0447)	(0.0424)	(0.0543)
Inequality ×	0.0787***	0.0801**	0.0764**	0.0877**	0.0543	0.1083***
Political competition	(0.0300)	(0.0325)	(0.0330)	(0.0399)	(0.0388)	(0.0380)
Panel C: MGNREGA (N=451,231):						
Inequality	-0.0139*	-0.0180**	-0.0145*	-0.0177*	-0.0176*	-0.0186*
	(0.0077)	(0.0085)	(0.0081)	(0.0100)	(0.0107)	(0.0102)
Political competition	-0.0034	-0.0020	-0.0030	0.0043	0.0048	0.0048
	(0.0095)	(0.0093)	(0.0093)	(0.0072)	(0.0055)	(0.0093)
Inequality ×	0.0104*	0.0103*	0.0126*	0.0088*	0.0008	0.0126*
Political competition	(0.0059)	(0.0057)	(0.0065)	(0.0052)	(0.0030)	(0.0068)

The independent variables are standardized, and the coefficients thus reflect the impact at average levels. The regressions control for average expenditures and area characteristics. In Panel A, we control for child characteristics as well. Panel A includes pre-delimitation constituency × district × birth year fixed effects, Panel B pre-delimitation constituency × district fixed effects, and Panel C pre-delimitation constituency × district × fiscal year fixed effects. Robust standard errors, clustered at the pre-delimitation constituency × district level, are shown in parentheses. *** significant at 1 percent, ** significant at 5 percent, * significant at 10 percent.

Figures

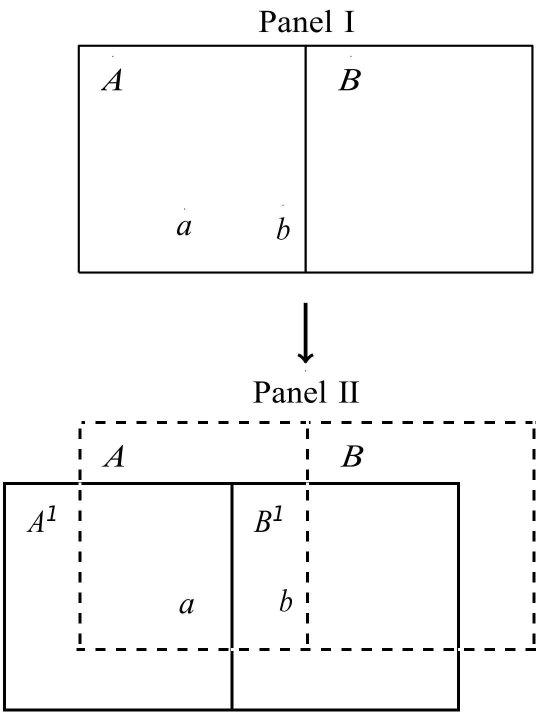

Figure 1: Illustration of the Identification Strategy

Notes

[1]The average infant mortality rate is estimated at 28 per 1,000 live births. For the worst performer, Madhya Pradesh, it is estimated at 43, and for the best performer, Mizoram, at 3.

[2]In our main specifications, we measure political competition as one minus the Herfindahl-Hirschman index and inequality using the Gini coefficient for household expenditures, but the effects are robust to different measures of inequality and political competition. Note also that households that change constituency could directly impact inequality and political competition. To deal with this, we calculate inequality and political competition before the boundary changes, using the borders of pre-delimitation constituencies. We then assign values to post-delimitation constituencies based on the pre-delimitation constituencies with the greatest population overlap.

[3]An independent Commission organized the Delimitation, and the consensus view is that it was carried out without great political influence (Bardhan et al. 2020; Iyer and Reddy 2013).

[4]In our main analysis, we focus on post-neonatal infant mortality (deaths between one and twelve months of age), which is primarily due to preventable diseases and malnutrition (Rai et al. 2017; Swaminathan et al. 2019). This type of mortality is much more likely to be affected by simple interventions than neonatal mortality (deaths before one month of age), which, to a larger extent, depends on the provision of individual clinical care (Lawn, Cousens and Zupan 2005).

[5] We can only study outcome variables for which information is available shortly after the Delimitation and which include geographical details (GPS coordinates) or information that allows us to link the data to the census.

[6]One exception is Fujiwara (2015), who studies a large enfranchisement of less-educated citizens in Brazil.

[7]When s is interpreted as the vote share, the HH index is calculated as $1 - (s^2 + (1 - s)^2) = 2s(1 - s)$.

[8]We calculate these numbers based on the "Question Hour Dataset" compiled by Bhogale (2019). Each question in the database contains a description of the subject. We searched through those subjects using the keywords "health", "employment", "unemployment" and "drinking water". In the Lok Sabha XIV (2004-2009), health is the subject for 547 questions (0.82%), employment for 487 questions (0.73%), unemployment for 61 questions (0.09%), and drinking water for 137 questions (0.21%). These numbers are roughly equal for the Lok Sabha XV (2009-2014).

[9]Given the lack of a thorough empirical investigation, we consider this analysis a separate contribution of our paper.

[10]Note that district borders did not change during the Delimitation.

[11]We do not have data on MGNREGA before the Delimitation, but we construct an alternative placebo test using the demand for work. We further detail this in Section V.B.

[12]Prior to the Delimitation, the correlation between political competition in 1999 and 2004 is 0.62. Comparing post-delimitation constituencies with their corresponding pre-delimitation constituencies, the correlation between political competition in 2004 and 2009 is 0.51. We plot this correlation in Online Appendix Figure D1. Moreover, the essence of proxies for political competition is to capture the likelihood, as perceived by the incumbent, that they will be ousted from office at the next election (Boyne 1998; Cronert and Nyman 2021; Immergut and Abou-Chadi 2014; Pettersson-Lidbom 2001). Prior election outcomes are likely to affect this perception, which also justifies our use of an earlier election outcome in the measure of political competition.

[13]This survey is the same as the Demographic and Health Survey (DHS).

[14]Mortality is the only health-related outcome in the NFHS for which we can construct such a retrospective time series, which is necessary to study the period shortly after the Delimitation.

[15]In our sample, there is only one candidate per political party among the ten largest candidates. Note that our results are robust to only including the five largest candidates in the calculation of the Herfindahl-Hirschman index.

[16]On average, the population share of the largest constituency within a district covers 77% of the district population. About 24% of the districts are located in one constituency only.

[17]This, therefore, assumes that expenditures are uniformly distributed across space within districts.

[18]For the relevant states, 71% of the households in the NFHS and 94% of the PHCs in the DLHS are located in rural areas. MGNREGA only covers rural areas, so the sample restriction does not apply to this outcome.

[19]These two sample restrictions reduce the mortality sample by 10%.

[20]For the MGNREGA sample, which is a panel over three years, we keep only one observation per GP. The mortality sample is not a panel, therefore we keep all children and the birth year in the fixed effects. The results are the same if we do not include the birth year and use θ'_{dk}, instead of θ'_{dkt}.

[21]We also test whether the observables are jointly significant. We do this by placing the binary variables, T_{idkl} and T_{jdkl}, on the left-hand side and all the listed observables, in addition to the fixed effects, on the right-hand side. The F-tests from this exercise are 0.76 for the mortality sample, 1.26 for the healthcare sample and 1.57 for the MGNREGA sample. This implies that our rich set of controls cannot predict which households, GPs and PHCs changed constituency.

[22]Table D1 shows a minor unbalance in religion, and it is a well-known puzzle that Muslim children are substantially more likely than Hindu children to survive their first birthday in India (see for example, Geruso and Spears 2018). Online Appendix C.1 shows the results are robust to the exclusion of the 10% of Muslim children, though.

[23]Online Appendix Table D3 presents the estimates for each of the variables underlying the healthcare index.

[24]Finding data on relevant outcome variables is challenging, as it has to cover the period shortly after the Delimitation and contain GPS coordinates or be mergeable with the census.

[25]Admittedly, the placebo tests are not perfect, as the discussion regarding the boundary changes started before 2008. We, therefore, cannot completely rule out that the post-delimitation constituencies played a role. It is reassuring, though, that we do not pick up any significant effects in these regressions.

[26]We calculate poverty rates using the NSS data and the so-called Tendulkar poverty lines, the latest set of official poverty lines in India (Government of India, 2012). The poverty lines reflect differences in the cost of living for each Indian state.

[27]We construct these fragmentation measures as Herfindahl-Hirschman indices of caste categories (STs, SCs and others) and religions (nine categories, measured by the NSS).

[28]There are only a few exceptions, all with a negative sign: In Panel A, the interactions of inequality with caste fragmentation in Column 5 (p-value: 0 .022), and with religious fragmentation in Column 6 (p-value 0.006); in Panel B, the interactions of inequality with mean expenditure in Column 1 (p-value: 0.059), and with literacy in Column 7 (p-value: 0.025); In Panel C, the interaction of inequality with caste fragmentation in Column 5 (p-value: 0 .039) and in Column 8 (p-value: 0 .099).

[29]The mean log deviation is calculated as $\frac{1}{N}\sum_{i=1}^{N}\log(\bar{y}\ /y_i)$, where i denotes households, y household expenditure, and $\bar{y}$ the arithmetic mean of household expenditures.

[30]The name is not entirely accurate in our setting, as we use the vote shares of candidates instead of parties. As mentioned before, there is only one candidate per political party. Due to independent candidates, the number of candidates is larger than the number of parties.

[31]Dash, Ferris and Winer (2019) argue that the MoV is an unreliable measure of political competition in India.

[32]The only exception is the impact on the healthcare index once we interact inequality and political competition with the party dummies.

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